Mason Bane

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Summary: Computer Science/Engineering student proficient across firmware (STM32, ARM Assembly, C/C++), hardware (circuit design, prototyping), and software (CUDA, OpenGL) domains. Combines embedded systems experience with high-performance computing research and complete system development skills. Strong analytical background with proven hardware/software integration capabilities.

Education

Tarleton State University Stephenville, TX
Bachelor of Science in Computer Science, Concentration in Computer Engineering *Aug. 2022 – May 2026*
Minor in Mathematics
GPA: 3.94/4.00 (Institutional) — Cumulative: 3.75/4.00

Hill College Hillsboro, TX
Associate of Arts in Liberal Arts *Aug. 2019 – Sept. 2023*

Experience

Undergraduate Research Assistant - Systems & Performance May 2024 – May 2026
Tarleton State University *Stephenville, TX*

- Developed and optimized high-performance computing simulations in C/C++, implementing parallel processing algorithms to maximize computational efficiency on Linux clusters.
- Engineered robust, maintainable software through systematic testing and debugging protocols, ensuring reliability for long-running computational workloads.
- Created comprehensive documentation for software deployment, configuration, and troubleshooting, facilitating collaboration and knowledge transfer across research teams.

Undergraduate Technology Specialist - HPC Lab Manager Aug. 2024 – May 2026
Tarleton State University *Stephenville, TX*

- Managed 15+ Linux systems in production-like research environment, ensuring 99%+ uptime through proactive maintenance, hardware diagnostics, and rapid incident response
- Diagnosed and resolved complex hardware/software issues including memory failures, storage degradation, and OS configuration problems, minimizing system downtime and maintaining optimal performance
- Developed and maintained comprehensive documentation for system configurations, troubleshooting procedures, and lab protocols, enabling knowledge sharing and process standardization

Projects

Secure IoT Sensor Node with Hardware TrustZone | *C, STM32CubeIDE/HAL, I2C/SPI* Dec. 2025 – May 2026

- Implementing a secure data pipeline on an STM32H5 MCU, utilizing Arm TrustZone and the Secure Manager to create a hardware-rooted chain of trust for environmental sensor data.
- Developing drivers and application logic to interface with a Bosch BME280 temperature, humidity, and pressure sensor via I2C communication protocol.
- Structuring firmware to cryptographically sign sensor readings within the secure processing environment for tamper-evident logging.
- Utilizing STM32CubeIDE and the HAL for peripheral configuration and hardware security module (HSM) management as a foundation for secure telemetry systems.

N-body Digital Twin of the Left Atrium | *NIH Grant #1R15HL179671-01* | *C, CUDA* Aug. 2024 – May 2026

- Engineered a parallel N-body model of the left atrium using CUDA, with a 20,000+ node mesh to simulate atrial arrhythmias in near real-time.
- Developed an intuitive C++/ImGui interface to control simulation parameters and visualize outputs, bridging the gap between complex computational models and end-user (research/clinical) requirements.
- Presented findings at academic conferences, demonstrating the tool's potential to improve clinical decision-making and transform training for medical professionals.

Technical Skills

Microcontroller Platforms: STM32 (H5/Cortex-M33), TIVA-C (TM4C), Arduino, Raspberry Pi, ARM Cortex-M
Firmware & Languages: C, C++, ARM Assembly, Python, Java, Bash, HTML/CSS/JavaScript, I^2C, SPI, UART
Hardware Tools: STM32CubeIDE, LTSpice, PCB Prototyping, Oscilloscope, Multimeter, GDB
Development & Systems: Git, Linux, CMake/Make, Visual Studio, VS Code, RTOS Concepts
Research & Simulation: MATLAB, CUDA, OpenGL, Blender, ImGui